AI101: Introduction to Artificial Intelligence • IIT Ropar

https://github.com/dk-iitrpr/fraudulent-payment-eda.git

Digital Payment Fraud Detection

A reproducible machine-learning pipeline for highly imbalanced financial transaction records using Logistic Regression with stratified evaluation and diagnostic ROC analysis.

Member Name	Entry No.	Role	GitHub Handle
Dikshant Kansal	2026EEB1345	Documentation Lead	@dk-iitrpr
Anish Meena	2026AIB1016	Run & Reproducibility Lead	@AnishMeena03
Nikita Patel	2026ESB1484	PPT Lead	@2026esb1484-cmd
Ayati Jain	2026MCB1520	Video Lead	@Ayati-itrpr

02 Problem and data

What and why

Digital financial transactions generate massive daily volume. Automated fraud detection catches illicit fund transfers in real time, preventing monetary loss and preserving platform trust.

Goal

Train and evaluate a supervised Logistic Regression classifier on financial features to reliably separate fraudulent cash-outs from legitimate transfers.

Dataset source

Online Payments Fraud Detection Dataset (Kaggle): Synthetic financial logs generated via PaySim with step, amount, balance shifts, and fraud labels.

6.36M
Total transactions (rows)
PaySim mobile money logs

11
Dataset features (cols)
5 selected numeric features

TARGET CLASS DISTRIBUTION (ISFRAUD)

Severe Imbalance (0.13%)

CLASS 0 • LEGITIMATE
6,354,407
99.87% majority baseline

CLASS 1 • FRAUDULENT
8,213
0.13% target anomaly

03 Approach & Architecture

Pipeline Mapped Directly to Repository Code

Entry Point: `python src/main.py` (executes the complete deterministic workflow in order)

1 • DATA LOADING

Raw Ingestion

Loads CSV records, verifies schema, isolates 5 numeric payment features + fraud label.

`src/data.py`
`load_data()`

pandas

2 • PREPARE & SPLIT

Preprocessing

80/20 stratified split. StandardScaler fitted exclusively on train fold to prevent leakage.

`src/data.py`
`split_data()`

scikit-learn

3 • MODELING

Logistic Classifier

L2-regularized logistic sigmoid model trained with max_iter=1000 and seed 42.

`src/model.py`
`build_model()`

scikit-learn

4 • EVALUATION

Metrics & Visuals

Computes ROC-AUC, confusion matrix, and exports metrics.json + diagnostic curves.

`src/evaluate.py`
`evaluate()`

matplotlib

src/config.py: Defines central dataset paths, target column `isFraud`, `TEST_SIZE=0.20`, and deterministic `SEED=42`.

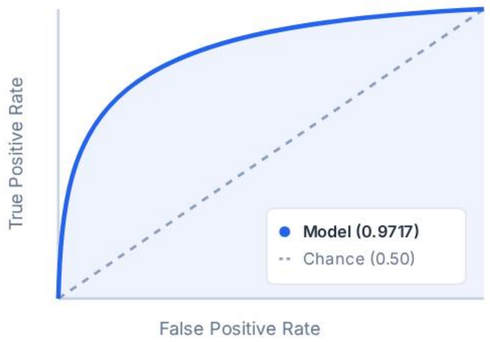
Python 3.10+ numpy pandas scikit-learn matplotlib

04 Results & Evaluation

Fresh-Clone Test Verified

ROC Curve

AUC = 0.9717



Confusion Matrix

Test: 1,272,524

TN (Legit Correct) 1,270,801	FP (False Alarm) 80
FN (Missed Fraud) 977	TP (Caught Fraud) 666

Execution Log Output

Passes Fresh Clone

```
$ python src/main.py
The shape of the dataset is (6362620, 11)
The column names of the dataset:
Index(['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrg', 'newbalanceOrig',
       'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud',
       'isFlaggedFraud'],
      dtype='str')
The head of the csv:
<bound method NDFrame.head of         step      type      amount  ...  newbalanceDest  isFraud  isF
laggedFraud
0          1   PAYMENT     9839.64  ...            0.00        0             0
1          1   PAYMENT     1864.28  ...            0.00        0             0
2          1  TRANSFER      181.00  ...            0.00        1             0
3          1  CASH_OUT      181.00  ...            0.00        1             0
4          1   PAYMENT    11668.14  ...            0.00        0             0
...      ...       ...         ...  ...             ...      ...           ...
6362615  743  CASH_OUT   339682.13  ...       339682.13        1             0
6362616  743  TRANSFER  6311409.28  ...            0.00        1             0
6362617  743  CASH_OUT  6311409.28  ...      6379898.11        1             0
6362618  743  TRANSFER   850002.52  ...            0.00        1             0
6362619  743  CASH_OUT   850002.52  ...      7360101.63        1             0

[6362620 rows x 11 columns]>

The needed csv details:
Features: ['amount', 'oldbalanceOrg', 'newbalanceOrig', 'oldbalanceDest', 'newbalanceDest']
Target: ['isFraud']
     amount  oldbalanceOrg  newbalanceOrig  oldbalanceDest  newbalanceDest  isFraud
0   9839.64       170136.0       160296.36            0.0             0.0        0
1   1864.28        21249.0        19384.72            0.0             0.0        0
2    181.00          181.0            0.00            0.0             0.0        1
3    181.00          181.0            0.00        21182.0             0.0        1
4  11668.14        41554.0        29885.86            0.0             0.0        0
C:\Users\erdik\miniconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array wa
s expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
        accuracy: 0.999
       precision: 0.893
          recall: 0.405
              f1: 0.558
         roc_auc: 0.972
baseline_accuracy: 0.999
The model is based on Logistic Regression & Accuracy over the model : 99.92 %
```

99.92%
Accuracy
Baseline: 99.87%

89.28%
Precision
Only 80 false alarms

40.54%
Recall
666 frauds detected

0.5576
F1-Score
ROC-AUC: 0.9717

05 Learnings & Responsibilities

All 4 Members Contributed Commits

What Worked Well

+ **Leakage Prevention:** StandardScaler fitted strictly on training data fold only.
+ **Path Portability:** Relative paths centralized in `src/config.py` runs from any directory.
+ **Baseline Defeated:** Achieved 99.92% accuracy, beating the 99.87% majority-class baseline.

What didn't work → Fixed

− **KeyError in load_data():** column list was passed as a tuple.
→ Fixed with a flat list of column names.

− **VS Code Interpreter:** libraries flagged as 'missing'.
→ Selected the correct local venv interpreter path.

Dikshant Kansal
Documentation lead
@dk-iitrpr

`README.md` `data.py` `repo structure` `evaluate.py`

Anish Meena
Execution lead
@AnishMeena03

`requirements.txt` `config.py` `fresh-clone test`

Nikita Patel
PPT lead
@2026esb1484-cmd

`5-slide deck` `model.py`

Ayati Jain
Video lead
@Ayati-itrpr

`5-minute video` `main.py`

AI-use disclosure Claude was used as a supporting tool during this project for understanding machine learning concepts and improving the code.